

A Packing of 10 Four-Spaces in $\mathbb{R}^9$

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Abstract

We give 10 4-dimensional subspaces of $\mathbb{R}^9$ whose smallest squared chordal distance is at least 2.468863. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.444132, obtained by deleting 1 of the 11 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 10 subspaces of dimension 4 in $\mathbb{R}^9$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.468863 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 9$, $n = 4$. The cell $N = 10$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.444132, obtained by deleting 1 of the 11 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(9 - 4)/9 \cdot 10/9 = 2.469136$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m9n4N10.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 61 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
617215754695784257549478766303799037302355368953654461175073411602264226
528062742833824543782235113623836321457506739982132205977662398597936363
014277955845956288575188368509646801088282686285
```

denominator:

```
250000000000185536789985775566364390901170871180855218692945537389679459
556982445009989330419308140677446329682658929894881654228815772857025163
629310373291780363932260894067984078174763824969
```

B Coordinates

10 blocks of 9 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m9n4N10.txt`.

```
96357167491 256331316013 398888109158 219543954043 274979771852 -20861944677 -141594343202 -386032630481 4104387077 -159750840061 232416022252 -304296081109
-33510342768 -435713898907 168454371687 79219417373 209999985969 -225431403291 -470244181303 -130494808858 746481662 -371934188964 723942687979 135337712913
187424059453 -291573585145 -720803393963 80993802112 225326259932 -368881149690 742062489017 -265405072370 148208064 -335679509819 -109276619545 -226721660547
714564065758 -354004480180 335588438551 -292223806547 4747963651 -635703262512 -247129954250 -84934822822 582473452 29953056826 -62537085483 -393652720485
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140438362324 539737558823 -110248670594 389665873634 11792369851 -104536239608 -116183843399 -373912303889 94755967 500550488503 224340840163 511323040992
5422671101 611521237346 269822221667 -590075156422 -184636753001 129997760249 93196398590 389022198878 172682745568 138060003411 -414718721557 185710919808
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```